Econ 814 Recitation 8 Notes

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1 Aug 2023 Comp Q2

Consider the following infinite horizon model. At the start of every period, a worker who is employed can continue to work at a constant wage, w. The worker has two options: either continue working at the current wage or quit, receive unemployment benefits b for the current period and engage in job search next time. If the worker decides to work, there is a probability ϕ that she will remain employed at wage w in the following period and a probability $1-\phi$ that she will become unemployed. An unemployed worker receives a wage offer from a distribution $F(w)$ and can choose to accept or reject the offer. Wage offers to unemployed workers remain independent and identically distributed (i.i.d.) over time.

The worker's preferences are $\sum \beta^t c_t$, where c_t is consumption. Assume no borrowing or lending. Also, if you make any additional assumptions, please state them explicitly.

1.1 Part a

Formulate the worker's decision problem using dynamic programming.

Solution: With no borrowing or lending by the optimality condition let $c_t = y_t$, that is the worker consumes whatever they receive in income.

In each period the worker must choose to maximize the choice of states between working and not working

$$V(w,e) = w + \beta \max_e \{V(w,e), V(w',u)\}$$

If the worker decides to remain employed

$$V(w,e) = w + \beta\phi V(u) + \beta(1-\phi)V(w,e)$$

Solve to find

$$V(w,e) = \frac{w + \beta\phi V(u)}{1-\beta(1-\phi)}$$

If the worker decides to quit, she faces unemployment benefits plus an uncertain wage in the future

$$V(w,u) = b + \beta \int_0^B V(w')dF(w')$$

So the workers problem is to maximize utility through maximizing the value of her state

$$V(w) = w + \max_e \{\frac{w + \beta\phi V(u)}{1 - \beta(1-\phi)}, b + \beta \int_0^B V(w')dF(w')\}$$

1.2 Part b

Characterize the reservation wage and show what happens to it as the level of b increases.

Solution: Since $V(w,u)$ is independent of w and $V(w,e)$ is increasing with w there exists a unique $\bar{w}$ satisfying

$$V(\bar{w}, e) = V(\bar{w}, u)$$

Where $\bar{w}$ is the reservation wage, the wage that makes the worker indifferent between accepting and rejecting an offer.

$$\frac{\bar{w} + \beta\phi V(u)}{1 - \beta(1-\phi)} = V(u)$$

Solving for $V(u)$ we find

$$\bar{w} + \beta\phi V(u) = (1 - \beta(1-\phi))V(u)$$

$$V(u) = \frac{\bar{w}}{1-\beta}$$

Breaking the integral into two pieces we have

$$\frac{\bar{w}}{1-\beta} = b + \beta \int_0^{\bar{w}} \frac{\bar{w}}{1-\beta} dF(w') + \beta \int_{\bar{w}}^B \frac{w' + \beta\phi\frac{\bar{w}}{1-\beta}}{1-\beta(1-\phi)} dF(w')$$

Break the left hand side into a factored integral since $1 = \int_0^{\bar{w}} dF(w') + \int_{\bar{w}}^B dF(w')$

$$\frac{\bar{w}}{1-\beta}\int_0^{\bar{w}} dF(w') + \frac{\bar{w}}{1-\beta}\int_{\bar{w}}^B dF(w') = b + \beta\int_0^{\bar{w}} \frac{\bar{w}}{1-\beta} dF(w') + \beta\int_{\bar{w}}^B \frac{w' + \beta\phi\frac{\bar{w}}{1-\beta}}{1-\beta(1-\phi)} dF(w')$$

Move the $\beta\int_0^{\bar{w}} \frac{\bar{w}}{1-\beta} dF(w')$ from the right hand side over to the left and factor

$$\frac{(1-\beta)\bar{w}}{1-\beta}\int_0^{\bar{w}} dF(w') + \frac{\bar{w}}{1-\beta}\int_{\bar{w}}^B dF(w') = b + \beta\int_{\bar{w}}^B \frac{w' + \beta\phi\frac{\bar{w}}{1-\beta}}{1-\beta(1-\phi)} dF(w')$$

Move the $\frac{\bar{w}}{1-\beta}\int_{\bar{w}}^{B} dF(w')$ from the left hand side over to the right hand side and factor

$$\bar{w}\int_{0}^{\bar{w}} dF(w') = b + \beta\int_{\bar{w}}^{B}[\frac{w' + \beta\phi\frac{\bar{w}}{1-\beta}}{1-\beta(1-\phi)} - \frac{\bar{w}}{\beta(1-\beta)}]dF(w')$$

Now add $\bar{w}\int_{\bar{w}}^{B} dF(w')$ to both sides

$$\bar{w} = b + \int_{\bar{w}}^{B}[\frac{w' + \beta\phi\frac{\bar{w}}{1-\beta}}{1-\beta(1-\phi)} - \frac{\bar{w}}{\beta(1-\beta)} + \frac{\bar{w}}{\beta}]dF(w')$$

Rearranging a bit

$$\bar{w} - b = \frac{\beta}{1-\beta(1-\phi)}\int_{\bar{w}}^{B}(w' - \bar{w})dF(w')$$

Where $\bar{w} - b$ is the cost of turning down wage offer $\bar{w}$ to continue searching and $\frac{\beta}{1-\beta(1-\phi)}\int_{\bar{w}}^{B}(w' - \bar{w})dF(w')$ is the expected discounted benefit of turning down a wage offer to continue searching. As the level of b increases, the cost for turning down the wage offer decreases, making it more enticing to stay out of the labor market. This leads to an increase in the reservation wage. You can see this formally by taking a derivative with respect to b.

$$\bar{w}(b) - b = h(\bar{w}(b))$$

$$\bar{w}'(b) - 1 = h'(\bar{w}(b))\bar{w}'(b)$$

$$\bar{w}(b) = \frac{1}{1 - h'(\bar{w}(b))}$$

To see that $h(\bar{w}(b)) < 1$, take the derivative using Leibnitz's rule:

$$g(x) = \int_{a(x)}^{b(x)} f(x,y)dy$$

$$g'(x) = f(x,b(x))b'(x) - f(x,a(x))a'(x) + \int_{a(x)}^{b}(x)\frac{\partial f(x,y)}{\partial x}dy$$

$$h(\bar{w}) = \frac{\beta}{1-\beta(1-\phi)}\int_{\bar{w}}^{B}(w' - \bar{w})dF(w')$$

$$h'(\bar{w}) = -\frac{\beta}{1-\beta(1-\phi)}(\bar{w} - \bar{w}) - \frac{\beta}{1-\beta(1-\phi)}\int_{\bar{w}}^{B} dF(w')$$

$$h'(\bar{w}) = -\frac{\beta}{1-\beta(1-\phi)}(1 - F(\bar{w})) < 0 < 1$$

1.3 Part c

Calculate the probability that an unemployed worker is unemployed for exactly N periods in a row as a function of the reservation wage and show that this probability is decreasing in N. What is the expected duration of unemployment?

Solution: $F(w)$ here is the cumulative probability of receiving a job offer at wage w. This means that the probability of receiving a job offer with a wage above the reservation wage (and thus accepting it) is $1 - F(\bar{w})$. The probability of not receiving a job offer with a wage above the reservation wage then would be $F(\bar{w}) = 1 - (1 - F(\bar{w}))$, and doing this for $N - 1$ periods before receiving an offer on the N period would be:

$$F(\bar{w})^{N-1}(1 - F(\bar{w}))$$

This is decreasing in N since F is by definition between 0 and 1 (a probability) with $F(0) = 0$ and $F(B) = 1$. The expected duration of unemployment then would be the sum of probabilities for each N starting from 1 to infinity.

$$\sum_{N=1}^{\infty} N * F(\bar{w})^{N-1}(1 - F(\bar{w}))$$

To solve this use the equation $\sum_0^\infty akr^k = \frac{ar}{(1-r)^2}$

$$\sum_{N=0}^{\infty} N * F(\bar{w})^N \frac{(1 - F(\bar{w}))}{F} = \frac{(1 - F(\bar{w}))}{F(\bar{w})} * \frac{F(\bar{w})}{(1 - F(\bar{w}))^2} = \frac{1}{1 - F(\bar{w})}$$

1.4 Part d

Suppose now that there are a large number of workers who all face the same problem as above. Define the unemployment rate as the fraction of all workers who are unemployed. How does the unemployment rate vary with b?

Solution: Define the number of unemployed people next period as Lu_{t+1}. This is equal to the number of unemployed people who do not accept a job offer plus the number of people employed who get fired

$$Lu_{t+1} = Lu_t F(\bar{w}) + L(1 - u_t)\phi$$

Simplifying to find the unemployment rate at steady state

$$u = uF(\bar{w}) + (1 - u)\phi$$

$$u = \frac{\phi}{\phi + (1 - F(\bar{w}))}$$

Here if the unemployment benefits increase, the reservation wage increases, which would raise $F(\bar{w})$, which would raise the unemployment rate.

2 Final 22 Q3

(15 points) Wage growth rate and the reservation wage: An unemployed worker receives each period an offer to work for wage w_t forever, where $w_t = w$ in the first period and $w_t = \phi^t w$ after t periods in the job. Assume $\phi > 1$, that is, wages increase with tenure. The initial wage offer is drawn from a distribution $F(w)$over $(0, B)$ that is constant over time (entry-level wages are stationary); successive drawings across periods are independently and identically distributed. The worker's objective function is to maximize

$$E \sum_{t=0}^{\infty} \beta^t y_t, \text{ where } \beta \in (0,1), \tag{1}$$

and $y_t = w_t$ if the worker is employed and $y_t = b$ if the worker is unemployed, where b is unemployment compensation. Let $v(w)$ be the optimal value of the objective function for an unemployed worker who has offer w in hand. Write Bellman's equation for this problem. Argue that, if two economies differ only in the growth rate of wages of employed workers, say $\phi_1 > \phi_2$, the economy with the higher growth rate has the smaller reservation wage. Assume that $\phi_i \beta < 1$, $i = 1; 2$.
Hint:
Write the reservation condition with a function of the reservation wage on the left-hand side and a function of the parameter ϕ on the left-hand side
You might then need the following rule

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(y,x)dy = f(b(x),x)\ b_x(x) - f(a(x),x)\ a_x(x) + \int_{a(x)}^{b(x)} f_x(y,x)dy$$

Although you do not need it, you might want to use the integration by parts formula:

$$\int_a^b udV = uV\ |_a^b - \int_a^b Vdu \tag{2}$$

Solution:

If the worker accepts employment at wage w, the sequence $\{y_t\}$ is given by $y_t = w$, $y_{t+1} = \phi w ... y_{t+j} = w\phi^t ...$. Therefore the value of the objective function if the worker accepts is

$$\sum_{j=0}^{\infty} \beta^j y_{t+j} = \frac{w}{1-\phi\beta}. \tag{3}$$

Bellman's equation for the worker's problem is

$$v(w) = \max_{A,R} \left\{ \frac{w}{1-\phi\beta}, b + \beta \int_0^B v(w')\, dF(w') \right\}$$

Using the same argument as when studying McCall's model, one shows that the optimal policy is to accept all offers to work with an initial wage higher than a

reservation wage $\bar{w}$:

$$v(w) = \begin{cases} \frac{\bar{w}}{1-\phi\beta} & \text{if } w \leq \bar{w} \\ \frac{w}{1-\phi\beta} & \text{if } w \geq \bar{w} \end{cases} \tag{4}$$

Because, at $w = \bar{w}$, we have

$$\frac{\bar{w}}{1-\phi\beta} = b + \beta \int_0^B v(w')\, dF(w') \tag{5}$$

Note that we can see how the LHS increases with ϕ which might lead us to believe that we can immediately infer that $\bar{w}$ will decrease so that the LHS stays equal to the RHS. However, the RHS also changes with $\bar{w}$ and ϕ, so we have to dig further.
After substituting for $v(w)$ the above expression becomes,

$$\begin{aligned} \frac{\bar{w}}{1-\phi\beta} &= b + \beta \int_0^{\bar{w}} \frac{\bar{w}}{1-\phi\beta} dF(w') + \beta \int_{\bar{w}}^B \frac{w'}{1-\phi\beta} dF(w') && (6) \\ &= b + \frac{\beta}{1-\phi\beta}\bar{w} \int_0^{\bar{w}} dF(w') + \frac{\beta}{1-\phi\beta} \int_{\bar{w}}^B w' dF(w') && (7) \\ &\Leftrightarrow && (8) \end{aligned}$$

$$\begin{aligned} \bar{w} &= b(1-\phi\beta) + \beta\bar{w} \int_0^{\bar{w}} dF(w') + \beta \int_{\bar{w}}^B w' dF(w') && (9) \\ &= b(1-\phi\beta) + \beta\bar{w} \int_0^{\bar{w}} dF(w') + \beta\bar{w} \int_{\bar{w}}^B dF(w') - \beta\bar{w} \int_{\bar{w}}^B dF(w') + \beta \int_{\bar{w}}^B w' dF(w') && (10) \\ &= b(1-\phi\beta) + \beta\bar{w} + \beta \int_{\bar{w}}^B (w' - \bar{w})\, dF(w') && (11) \\ &\Leftrightarrow && (12) \end{aligned}$$

$$\begin{aligned} (1-\beta)\bar{w} - \beta \int_{\bar{w}}^B (w' - \bar{w})\, dF(w') &= (1-\phi\beta)\, b && (13) \\ &\Leftrightarrow \text{ (integrating by parts, see notes)} && (14) \\ (1-\beta)\bar{w} - \beta \int_{\bar{w}}^B (1 - F(w'))\, dw' &= (1-\phi\beta)\, b && (15) \end{aligned}$$

It is easy to see (using the rule provided) that

$$\frac{\partial \left[(1-\beta)\bar{w} - \beta \int_{\bar{w}}^B (w' - \bar{w})\, dF(w') \right]}{\partial \bar{w}} = (1-\beta) - \beta \frac{\partial \left[\int_{\bar{w}}^B (w' - \bar{w})\, dF(w') \right]}{\partial \bar{w}} \tag{16}$$

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(y,x)dy = f(b(x),x)\ b_x(x) - f(a(x),x)\ a_x(x) + \int_{a(x)}^{b(x)} f_x(y,x)dy$$

So

$$\frac{\partial\left[\int_{\bar{w}}^{B}\left(w'-\bar{w}\right)dF\left(w'\right)\right]}{\partial\bar{w}} = \left(B-\bar{w}\right)\times 0-\left(\bar{w}-\bar{w}\right)\times 1+\int_{\bar{w}}^{B}\left(-1\right)dF\left(w'\right) \quad (17)$$

$$= -\left[F\left(B\right)-F\left(\bar{w}\right)\right] \quad (18)$$

$$= -\left[1-F\left(\bar{w}\right)\right] \quad (19)$$

$$S'(w) = (1-F(B))\cdot 0-(1-F(w))\cdot 1+\int_{w}^{B}0.dw'$$

the left-hand side is increasing in $\bar{w}$. Therefore, if $\phi_1 > \phi_2$, that is, $(1-\phi_1\beta) < (1-\phi_2\beta)$, it must be that $\bar{w}_1 < \bar{w}_2$. The intuition behind this result is simple: for any given offer w, the value of accepting the offer is higher, the higher the growth rate of wages ϕ. Therefore, the sooner an offer is accepted, the sooner the benefits of the growth in wages are realized. This pattern makes some job offers more attractive even though the initial wage is not very high.